

2. Introduction

Spin qubits in gate-defined semiconductor quantum dots are a promising platform for quantum computing, owing to their potential for scaling and their compatibility with established semiconductor fabrication technology.^{1–3)} In these devices, quantum information is encoded in the spin of electrons confined by electrostatic potentials, and the confinement is defined by the voltages applied to an array of gate electrodes. Operating a device therefore requires that these voltages be set such that each dot holds a precisely known number of electrons, typically one electron per dot. As devices scale to larger arrays, the number of gate voltages to be set grows accordingly, and determining the electron occupation efficiently becomes a central experimental challenge.

The electron occupation is determined from the charge stability diagram (CSD).⁴⁾ A CSD is acquired by sweeping two gate voltages while recording the signal of a capacitively coupled charge sensor. Each change in the occupation of a dot shifts the electrostatic potential at the sensor and produces an abrupt change in its signal. The resulting two-dimensional map consists of regions of fixed charge occupation separated by sharp charge-transition lines, which for a double quantum dot form a honeycomb pattern. Two families of lines correspond to the loading of an electron from a reservoir into either dot, and short interdot lines to the transfer of an electron between the dots. Because each dot couples more strongly to its own plunger gate than to the other, the loading lines of the two dots have markedly different slopes in the (V_1, V_2) plane. Identifying these lines, and counting them from the fully depleted $(0,0)$ configuration, establishes the electron occupation (N_1, N_2) throughout the diagram.

Most of the CSD, however, carries no information about where the transitions occur. Within each honeycomb cell the dots are in Coulomb blockade: the charging energy prevents the addition of an electron, so the occupation (N_1, N_2) is constant throughout the cell and the sensor signal varies only smoothly. The information that defines the occupation is concentrated on the transition lines, which occupy only a small fraction of the voltage plane. In practice, however, a CSD is measured point by point on a regular two-dimensional grid: for each value of V_2 , the gate voltage V_1 is swept across its full range, so that every pixel of the voltage plane is measured, including those within Coulomb-blockaded regions. The number of measurements therefore grows quadratically with the resolution, and the large majority of them sample regions whose occupation could be inferred from the surrounding lines.

Hernandes et al. have recently addressed the cost of this acquisition with a generative approach.⁵⁾ They trained a conditional diffusion model on approximately 9000 measured CSDs to reconstruct full diagrams from sparse measurements. Two sampling strategies were evaluated: a uniform grid of points, from which the transition lines were preserved with as little as 4% of the data measured, and horizontal and vertical line cuts, evaluated at 23–44% of the data. The model reconstructs the sensor signal over the entire plane, including the featureless Coulomb-blockaded regions, and its output is an image. The positions of the transition lines, which are the quantity that determines the electron occupation, are extracted from the reconstructed image only afterwards, using classical edge- and ridge-detection filters. The same filters, applied to the measured diagrams, provide the reference lines against which the reconstructions are evaluated.⁵⁾ Both the training data and the reference transition lines therefore derive from measurement. The number of available diagrams is bounded by the experimental effort, the dataset had to be curated by manually labelling a subset of diagrams by quality, and the reference lines are only as accurate as the detectors that extract them.

Recent advances in the simulation of quantum-dot devices have made training data of this kind inexpensive to generate. Fast solvers of the constant-capacitance model are now available: QArray computes a 100×100 pixel charge stability diagram of a 16-dot array in less than a second, and smaller arrays in milliseconds, faster than they could be measured experimentally.⁶⁾ Related packages include QDsim⁷⁾ and QDarts, which finds charge transitions in the presence of finite tunnel couplings, non-constant charging energies, and sensor dots.⁸⁾ Most recently, QArray+ has extended this framework into a physics-informed simulator.⁹⁾ It includes the coherent tunnel coupling between neighbouring dots, whose strength depends on the gate voltages. Near an interdot transition, where two charge configurations are nearly degenerate, the ground state becomes a superposition of both. The simulated diagrams therefore exhibit the avoided crossings and fractional charge occupations observed in tunnel-coupled devices.⁹⁾

The simulator stands in for the device while the network is trained, supplying both the stability diagrams and the transition lines that label them; the network is given only what a fan of rays measured, and returns a probability map for a device it was not trained on.

We therefore propose to train a neural network entirely on simulated diagrams and to employ it in experiments as a probabilistic model of the transition-line positions. Given only a small number of one-dimensional rays measured on a device, the network returns, for every pixel of the voltage plane, the probability that a charge-transition line passes through it. It

does so without reconstructing the sensor signal. The rays form a fan of oblique sweeps emitted from one corner of the gate-voltage window, along each of which both gate voltages change simultaneously.¹⁰⁾ A horizontal line cut runs nearly parallel to the loading lines of one dot and a vertical cut to those of the other, so that each axis-aligned cut predominantly samples a single family of lines. Each oblique ray, in contrast, intersects both families. The network is a U-Net that receives the ray data as two image channels. The first contains the sensor signal at the sampled pixels. The second is a binary mask of the sampled pixels, which allows the network to distinguish a sampled pixel within a Coulomb-blockaded region from a pixel that was not sampled. In the present work, the training devices are simulated with QArray.

Three methodological choices are stated explicitly, since the reported results depend upon them. First, the measurement budget is quantified as the fraction of the voltage plane actually sampled by the rays, rather than as a number of measurement points. Second, the threshold that converts the probability map into binary transition lines is selected on a validation subset drawn from the training devices, never on the held-out test devices, and it is reported with each result. Third, because the transition lines are one pixel wide, a strict pixel-overlap metric treats a displacement of a single pixel as a complete miss. We therefore adopt a tolerance-based F1 score, $F1@τ$, and report it throughout together with the tolerance $τ$ and with the fraction of the plane lying within $τ$ pixels of a sampled pixel. The strict intersection-over-union is reported alongside it.

With eight rays of 60 points each, sampling 4.7% of the voltage plane, the predicted transition lines reach $F1@1 = 0.82$ on 50 held-out devices. For reference, 18.9% of the plane lies within one pixel of a sampled pixel.

In this study, both training and evaluation are performed on devices simulated within the constant-capacitance model, with capacitances drawn independently at random for each device, and the simulations are free of noise. The result established here is therefore the first step of the proposed approach: the charge-transition lines of a double quantum dot can be located from rays sampling a few percent of the voltage plane. Deployment of the trained model on measured devices, whose diagrams are subject to sensor noise, charge switching, and deviations from the constant-interaction picture, remains the subject of future work.

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